THE UNITED REPUBLIC OF TANZANIA
PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
NJOMBE TOWN COUNCIL

FORM FOUR PRE- MOCK EXAMINATION APRIL -2024

031/1 **PHYSICS 1**

TIME: 3:00 HOURS **25TH APRIL 2024**

INSTRUCTIONS

1. This paper consists of sections A, B and C
2. Answer all questions in all sections.
3. All writings should be in blue/black ink EXCEPT drawings if any which must be in pencil.
4. Calculators and cellular phones are not allowed in the examination room.
5. Write your index number on every page of your answer sheet.
6. Where necessary the following constants may be used.

(i) Acceleration due to gravity, g =10m/s^2 or 10 N/kg.
(ii) Density of water = 1g/cm^3 or 1000kg/m^3.
(iii) Speed of light in air=3×10^8m/s
(iv) 1h.p=746W

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SECTION A (16 marks)
Answer all questions from this section

1. For each of the items (i)-(x) choose the correct answer from among the given alternatives then write the letter of the correct answer in the answer sheets provided.

i. A metal X and a metal Y both loses the same quantity of heat when their temperature falls from 20^0C to 10^0C. the specific heat capacity of X is twice that of Y. then the ratio mass of X/mass of Y is;

A. 3/1 C. 1/2 E. 1/4
B. 2/1 D. 1/3

ii. A vacuum thermos flask is used to store some liquid nitrogen at -160^0C. the inside walls of the vacuum jacket are silvered so that heat is not;

A. Conducted toward the liquid
B. Radiated towards the liquid
C. Gained by convection
D. Lost by convection
E. Radiated away from the liquid

iii. The magnetic field plotted around a closed, small box X on a horizontal table is a pattern of circle round X with the same centre. The inside X is;

A. A bar magnet
B. A current carrying solenoid
C. A soft iron bar
D. A horizontal current carrying straight wire
E. A vertical current carrying straight wire

iv. When a lighted match is brought near a negatively charged electroscope, the leaf of the electroscope collapses because the;

A. Heat increases molecular vibrations which knock of the electrons from the cap
B. Flame provides energy for the electrons to escape from the cap
C. Match emits gamma rays
D. Positive charge from the air ionized by the flame are attracted by the cap
E. Match emits α-particles

v. An auto-mechanic wants to measure the diameter of a wire to the nearest o.oo1cm. Which of the following instruments can be used?

A. Vernier caliper D. Pair of compasses
B. Micrometer screw gauge E. Engineer's caliper
C. Metre rule

vi. A lens is used to form an image of a bright object on a screen. The effect of covering the top half of the lens with a card is to;

A. Make the image less bright
B. Make the image go out of focus
C. Make the image disappear
D. Remove the top half of the image
E. Remove the bottom half of the image

vii. A gas of volume 900cm^3 at 27^{0}C when warmed at a constant pressure to 87^{0}C will occupy a new volume of;

A. 900cm^3
B. 720cm^3
C. 1080 cm^3
D. 540 cm^3
E. 727 cm^3

viii. Which of the following statements is true when the resistance, R of a wire is measured using an ammeter, voltmeter and rheostat?

A. The ammeter is in parallel with R
B. The voltmeter is in series with R
C. A graph of V against I has a gradient equal to R
D. A graph of I against V has a gradient equal to R
E. The rheostat is in parallel with R

ix. The destruction of ozone layer in the atmosphere has led the world disasters caused by;

A. Absorption of the background radiation
B. Poor handling of the radioactive materials
C. The increase of the ultra-violet radiations on the earth
D. Poor utilization of radioactive materials
E. Poor disposal methods of radioactive remains

x.

xi. In a race competition, one should kneel down when starting the race. What is the reason of doing this;

A. No one can fall down by reaction produced by the earth to them
B. Each one can receive an acceleration which the earth provides
C. One of the rules in sports is not violated by anyone
D. Equal and opposite forces can act on a body to produce resultant force
E. Each one can exert a force on the surface of the ground

2. Match the items in List A with response in the List B by writing the correct response beside the item in the answer sheet provided.

LIST A	**LIST B**
i. The boundary that separate stratosphere from other layers	A. Stratosphere
ii. The region found in exosphere where satellites orbit the earth	B. Ionosphere
iii. The collective name given to troposphere and stratosphere	C. Magnetosphere
iv. The region nearest the earth of which most weather phenomena occur	D. Stratopause
v. The layer in which the ozone layer is found	E. Tropopause
vi. The layer in which most meteor burn while entering the earth's atmosphere.	F. Magnetosphere
	G. Upper atmosphere
	H. Lower atmosphere
	I. Mesosphere
	J. Troposphere
	K. thermosphere

SECTION B: (54 MARKS)

Answer all questions

3. (a) People living in areas that are said to be associated with the occurrence of earthquakes are notified through indicators. Suggest three indicators of earthquakes
   (b) The temperature of the atmosphere increase day by day as it is caused by greenhouse gases. Discuss four main sources of greenhouse gases.

4. (a) Describe and explain how a full wave rectification can be achieved using two diodes.
   (b) A tin contains water at 290K and is heated at constant rate. It is observed that the water reaches boing point after 2minutes and after further 14 minutes it is completely boiled away. Calculate the specific latent heat of steam.

5. (a) The half-life of a certain element is 64 days. What is the meaning of this statement?
   (b)A radioactive element has a half-life of 3min. if the initial count rate is 512 per min;
   (i) How long does it take to reach a count rate of 64 per min?
   (ii) What fraction of the original number of atoms is then left?

6. (a) The shortest length of the air column in a resonance tube (with one end closed and one end open) which resonates to a note of frequency 600Hz is found to be 170mm. what is the shortest length of the column of air which resonates in similar conditions to a note of frequency 900Hz?

   (b) Explain in terms of sound waves, why a piano can be distinguished from a violin even when they are playing notes of the same pitch?

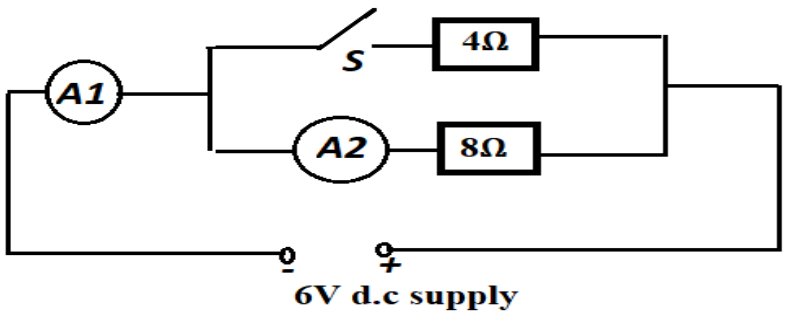
7. (a) A metal pipe which is 2m long at 30^0C increases in length by 0.5% when carrying a steam at 100^0C. Find the coefficient of expansion of the metal.
   (b) With the aid of diagram explain how the use of a vacuum flask reduces heat transfer by conduction, convection and radiation.

8. (a) Why does a transformer work with alternating current (a.c) only?
   (b) A calorimeter of heat capacity 42J/K contains $500cm^3$ of water at 20^0C. An automatic stirrer dips into the water and is driven by a 1/4h.p motor. Assuming that all the work done by the stirrer heats the water. Calculate the temperature of the calorimeter and its contents after 10min.

SECTION C: 30 MARKS

Answer only two questions

9. (a)(i) The name dry cell means that the cell has no liquid inside it. Draw a well labelled diagram of the dry cell
   (ii) Suggest two defects of a simple cell
   (b) in the figure below, the resistances of ammeters may be ignored. Explain each of the following observations;



(i) When switch S is closed, the current through ammeter 2 (A2) is less than that through ammeter 1 (A1)

(ii) When switch S is open, the current through ammeter1(A1) falls

(iii) When switch S is open, the current through each ammeter is the same.

10. (a) In the induction coil, the induced e.m.f is very large and is usually in the order of hundreds of kilovolts (kV). Briefly explain two ways in which such a high voltage can be achieved.

    (b) With the aid of diagram describe the structure and operation of a d.c generator

    (c) A current of 0.7A is passed through a step down transformer with a primary coil of 500turns. A current of 0.2A is obtained in the secondary coil. Determine the number of turns in the secondary coil and the voltage across if the primary coil is connected to a 240V mains.

11. (a)`In a light experiment, a narrow beam of light directed onto a glass prism leaves the prism and falls on a white screen. Draw a labelled diagram to show the experiment set-up and observation seen on screen.

    (b) A screen is placed 60cm from an object. A lens is used to produce on the screen an image with magnification of 4. Calculate;

    (i) Distance between the object and the lens

    (ii) Focal length of the length

    (c) Explain two ways in which lens cameras differ from human eye.